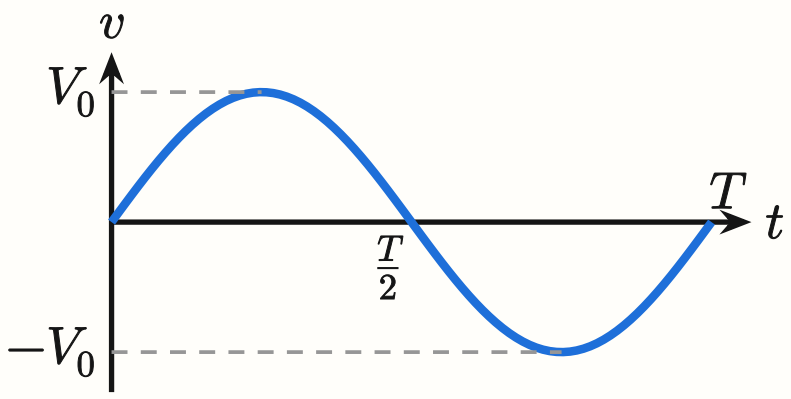
Alternating Current

Class 12 · SINUSOIDAL source, STEADY state · IDEAL L, C (no resistance) unless stated · $v = V_0 \sin \omega t$, $i = I_0 \sin(\omega t - \varphi)$: $\varphi > 0$ current LAGS · V, I = rms values unless stated

1 Alternating voltage



$$v = V_0 \sin \omega t$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

v = instantaneous value, V_0 = peak value (amplitude) · ω in $\mathrm{rad\,s^{-1}}$, f in Hz, T = period · current: $i = I_0 \sin(\omega t - \varphi)$

2 Why AC

- An AC voltage is stepped up or down easily by a transformer (it needs a CHANGING current).
- Power is sent at HIGH voltage and LOW current, so the line loss I^2R is small.

Example

Household mains (India), $220\,\mathrm{V}$ rms, $50\,\mathrm{Hz}$:
$V_0 = 220\sqrt{2} = 311.1\,\mathrm{V}$, $\omega = 2\pi \times 50 = 314.16\,\mathrm{rad\,s^{-1}}$, $T = 0.02\,\mathrm{s}$.

3 DC vs AC

- DC: one direction (ideal battery: constant). Sinusoidal AC passes through zero twice per cycle: it REVERSES $2f$ times per second (100 times at $50\,\mathrm{Hz}$).

Syllabus Checklist

JEE Main + JEE Advanced + NEET syllabus + standard results

1 JEE Main + JEE Advanced syllabus

- ✓ Alternating current and voltage; AC vs DC — p1
- ✓ Peak, mean and rms values (sine) — p3
- ✓ Pure R, L, C circuits: phase, reactance — p7–p9
- ✓ Reactance and impedance — p8–p12
- ✓ Series LR, RC, LCR circuits (phasor diagrams) — p10–p13
- ✓ Resonance, Q-factor, bandwidth, sharpness — p14, p15
- ✓ Power in AC, power factor, wattless current — p17, p18
- ✓ LC oscillations (energy, analogy with SHM) — p20–p22
- ✓ AC generator and transformer — p23–p25

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2 Also covered: hidden topics + standard results

- ✓ Phasors and adding phasors — p6
- ✓ Mean and rms of other waveforms; DC + AC — p4
- ✓ AC meters: hot-wire (rms) vs moving-coil (mean) — p5
- ✓ Half-power frequencies (exact), voltage across L, C — p15
- ✓ Series LC; parallel LC and LCR resonance — p16
- ✓ Apparent, true and reactive power — p18
- ✓ Choke coil — p19
- ✓ Transformer efficiency, losses, transmission — p25, p26
- ✓ DC steady state in L, C circuits — p27
- ✓ Special cases, results, traps, solved questions — p27–p35

Mean and RMS Values

$v = V_0 \sin \omega t$ **(sinusoidal)**, $T = \dfrac{2\pi}{\omega}$

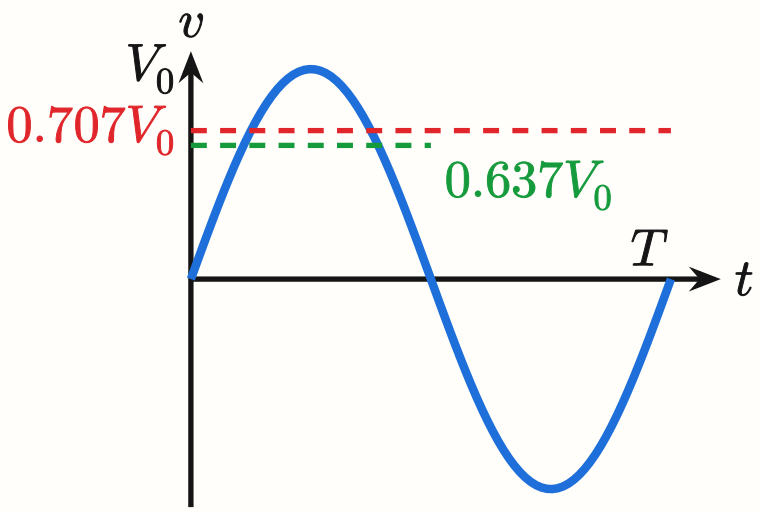
1 Mean (average) value

$$\bar{v} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} v \, dt$$

$$\bar{v}_{half} = \frac{2V_0}{\pi} \approx 0.637 V_0 \qquad \bar{v}_{full} = 0$$

half: over a POSITIVE half cycle, $t = 0$ to $\frac{T}{2}$ · full: one COMPLETE cycle (areas cancel)

2 Root mean square (rms) value



$$V_{rms} = \sqrt{\overline{v^2}} = \frac{V_0}{\sqrt{2}}$$

$\approx 0.707 V_0$ · *SINUSOIDAL only* · $\overline{\sin^2 \omega t} = \frac{1}{2}$ *over a full (or half) cycle* · *same for current:* $I_{rms} = \frac{I_0}{\sqrt{2}}$

- *rms = the steady (DC) value giving the SAME heat in the same resistor in the same time. AC meters and ratings (220 V) are rms.*

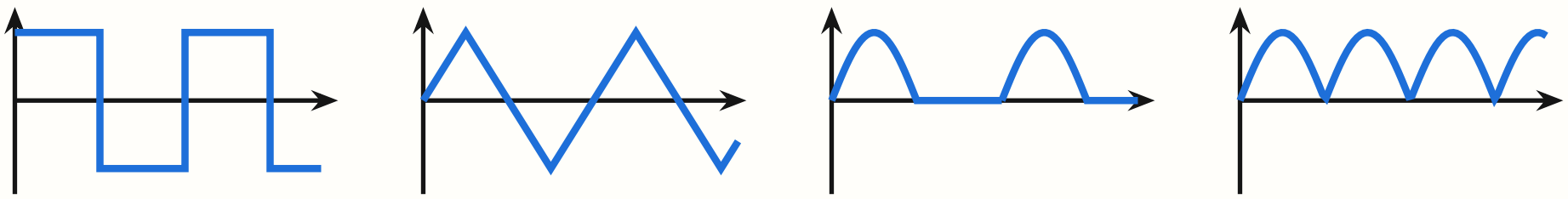
Example

$I_0 = 10\ \text{A}$: $I_{rms} = 7.07\ \text{A}$; *mean:* $6.37\ \text{A}$ *(positive half cycle),* 0 *(full cycle).*

RMS of Other Waveforms

peak value V_0, period T; mean = average over the stated interval

1 Four common shapes



square $\pm V_0$ | triangular $\pm V_0$ | half-wave rectified | full-wave rectified

2 Table

Waveform	rms	mean, full period	mean, positive half
sine	$\frac{V_0}{\sqrt{2}}$	0	$\frac{2V_0}{\pi}$
square $\pm V_0$	V_0	0	V_0
triangular $\pm V_0$	$\frac{V_0}{\sqrt{3}}$	0	$\frac{V_0}{2}$
half-wave rectified sine	$\frac{V_0}{2}$	$\frac{V_0}{\pi}$	$\frac{2V_0}{\pi}$
full-wave rectified sine	$\frac{V_0}{\sqrt{2}}$	$\frac{2V_0}{\pi}$	$\frac{2V_0}{\pi}$

3 DC + AC together

$$i = I_1 + I_2 \sin \omega t \;\Rightarrow\; I_{rms} = \sqrt{I_1^2 + \frac{I_2^2}{2}}$$

cross term $2I_1 I_2 \sin \omega t$ averages to zero over a full cycle.
$i = I_1 \sin \omega t + I_2 \cos \omega t$ (same ω): $I_{rms} = \sqrt{(I_1^2 + I_2^2)/2}$

AC Meters and Heating

1 Heating effect

$$P_{avg} = I_{rms}^2 R$$

resistor R, ANY periodic current (defines rms) · sinusoidal: $P_{avg} = \frac{1}{2} I_0^2 R$

2 Which meter reads what

Meter	deflection $\propto$	works on	reads
moving-coil (permanent magnet)	current i	DC	MEAN: 0 for sinusoidal AC
hot-wire	i^2 (heating)	AC and DC	rms
moving-iron	i^2	AC and DC	rms

At $50\,\mathrm{Hz}$ the moving-coil pointer cannot follow the reversals: it shows the MEAN (zero for a symmetric AC). Meters $\propto i^2$ read the same for a DC and an AC of equal rms value.

3 Form factor

$$\text{form factor} = \frac{I_{rms}}{\bar{I}_{half}} = \frac{\pi}{2\sqrt{2}} \approx 1.11$$

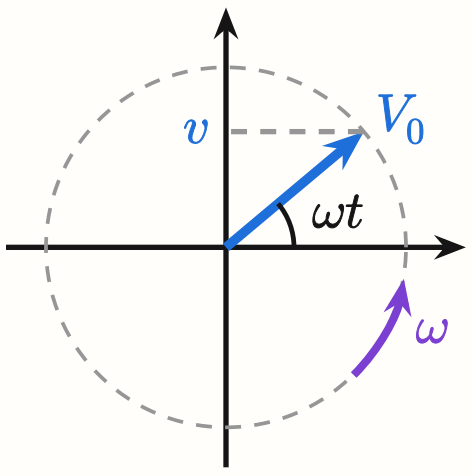
SINUSOIDAL · $\bar{I}_{half}$ = mean over a positive half cycle

Example

A moving-coil and a hot-wire ammeter in series carry $i = 2 + 3\sin\omega t$ A: moving-coil reads the mean, $2\,\mathrm{A}$; hot-wire reads the rms, $\sqrt{4 + 4.5} = 2.92\,\mathrm{A}$.

Phasors

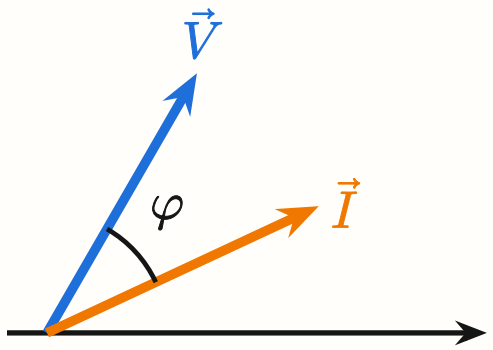
1 Phasor = rotating vector



$$v = V_0 \sin \omega t$$

= projection of a vector of length V_0, at angle ωt to the horizontal, on the VERTICAL axis · it rotates ANTICLOCKWISE at ω · all phasors of one circuit have the same ω, so the angles BETWEEN them stay fixed

2 Phase difference



- Current LAGS by φ: $\vec{I}$ is φ BEHIND $\vec{V}$ (clockwise from it). Leading: AHEAD (anticlockwise).
- Diagrams usually use rms phasors ($V_0/\sqrt{2}$): same angles.

3 Adding phasors

$$V_0 = \sqrt{V_1^2 + V_2^2 + 2V_1V_2 \cos \delta}$$

two SINUSOIDAL voltages of the SAME ω, amplitudes V_1, V_2, phase difference δ: add like vectors

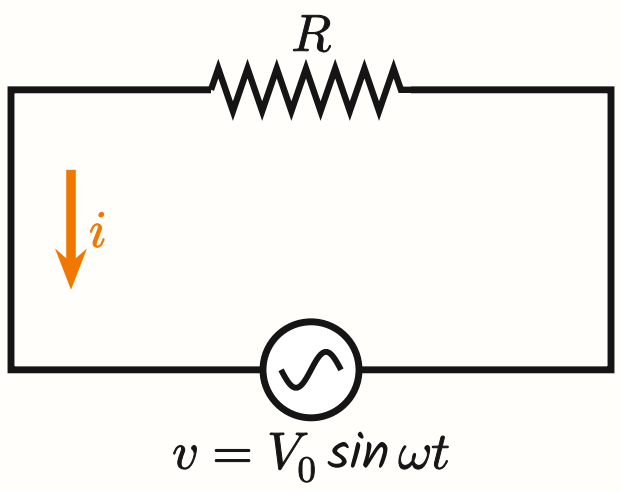
Example

$v_1 = 3 \sin \omega t$, $v_2 = 4 \cos \omega t = 4 \sin(\omega t + 90°)$ (volts):
$v_1 + v_2 = 5 \sin(\omega t + 53.13°)$, since $\tan \alpha = \frac{4}{3}$.

Pure Resistor

source $v = V_0 \sin \omega t$ across an ideal resistor R only

1 Circuit and current

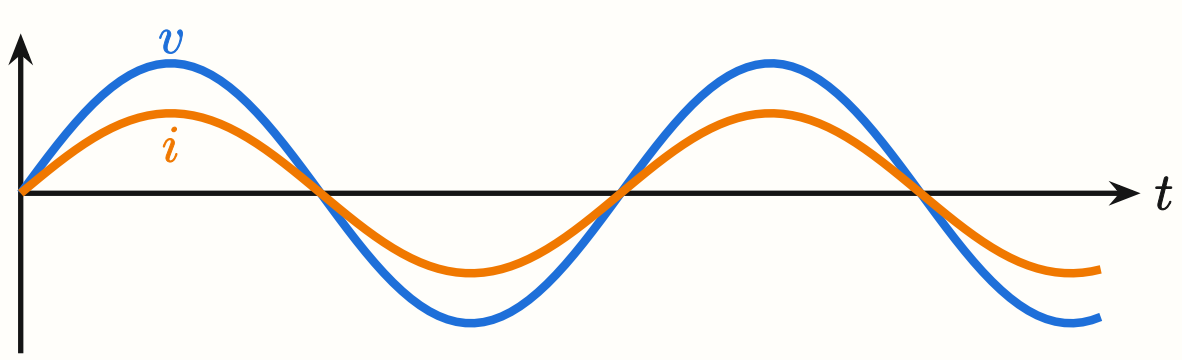


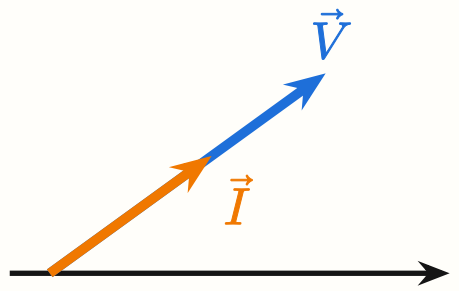
$$i = \frac{V_0}{R} \sin \omega t$$

$$I_{rms} = \frac{V_{rms}}{R}$$

IN PHASE ($\varphi = 0$) · Ohm's law at every instant · R independent of f

2 Graph and phasor





Peaks and zeros together: phase difference zero; phasors along the same line.

3 Power

$$P = V_{rms} I_{rms} = I_{rms}^2 R = \frac{V_{rms}^2}{R}$$

average over a cycle · power factor $\cos 0 = 1$

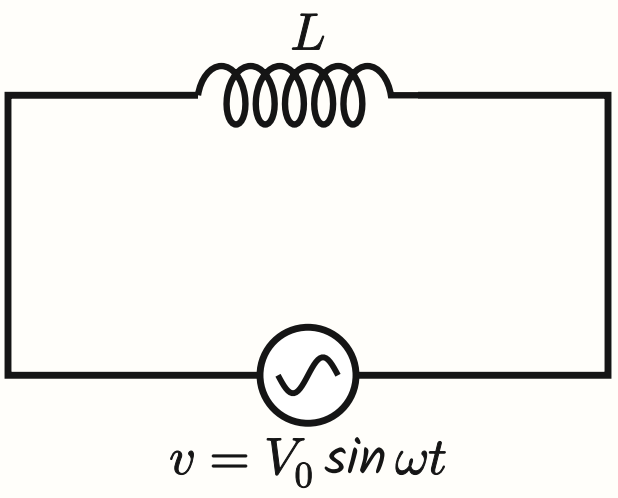
Example

$220\,\mathrm{V}$ *rms across* $110\,\Omega$: $I_{rms} = 2\,\mathrm{A}$, $I_0 = 2.83\,\mathrm{A}$, $P = 440\,\mathrm{W}$.

Pure Inductor

source $v = V_0 \sin \omega t$ across an IDEAL inductor L only (no resistance)

1 Circuit and current



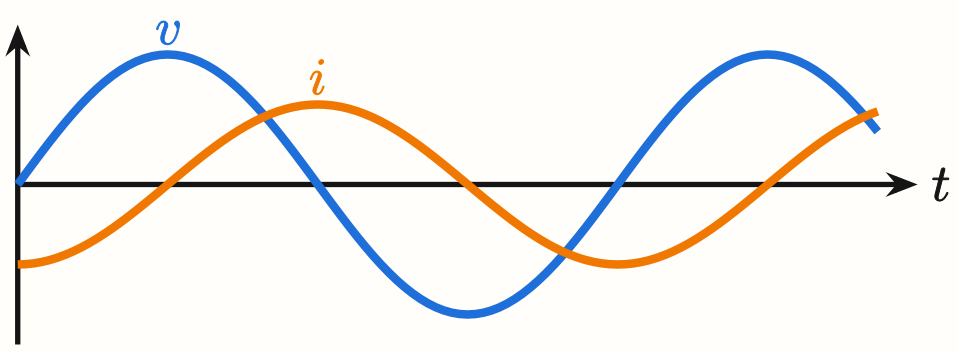
$$V_0 \sin \omega t = L\frac{di}{dt}$$

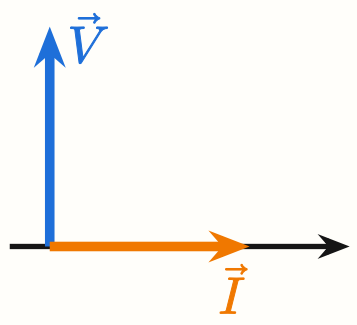
$$i = I_0 \sin\left(\omega t - \tfrac{\pi}{2}\right)$$

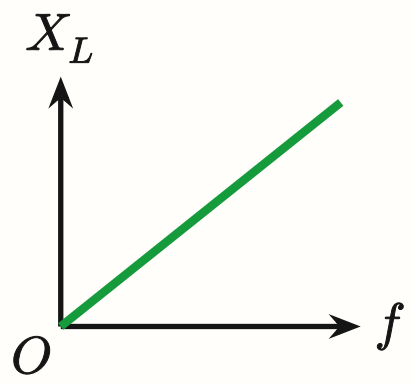
$$X_L = \omega L = 2\pi f L, \quad I_0 = \frac{V_0}{X_L}$$

current LAGS by $\frac{\pi}{2}$ · X_L: inductive reactance

2 Graphs and phasor







- i peaks a quarter period AFTER v. $X_L \propto f$; DC (steady, $f = 0$): $X_L = 0$, a plain wire.
- Average power $= 0$: field energy stored in one quarter cycle is returned in the next.

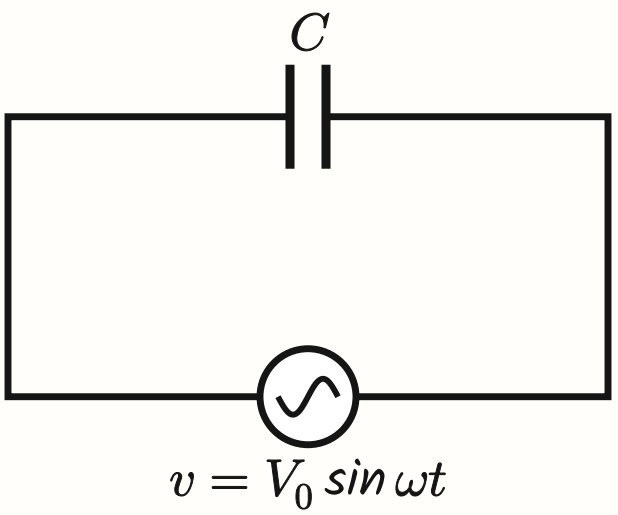
Example

$L = 0.2\,\text{H}$ on $220\,\text{V}$, $50\,\text{Hz}$: $X_L = 62.83\,\Omega$, $I_{rms} = \frac{220}{62.83} = 3.50\,\text{A}$.

Pure Capacitor

source $v = V_0 \sin \omega t$ across an IDEAL capacitor C only

1 Circuit and current



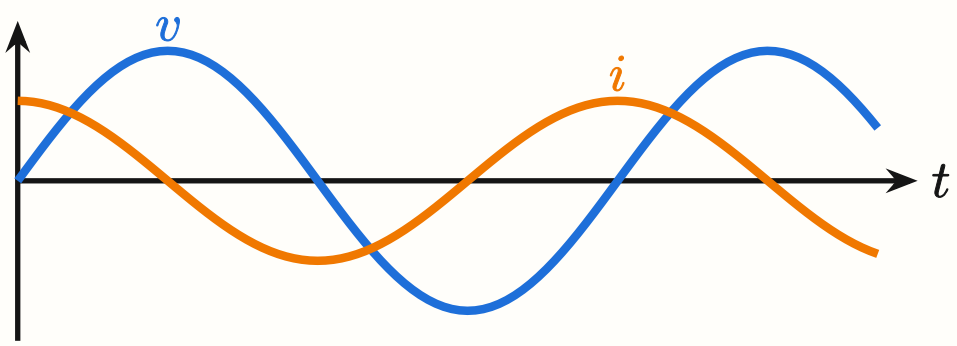
$$i = \frac{dq}{dt} = C\frac{dv}{dt}$$

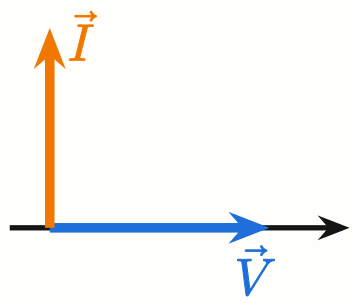
$$i = I_0 \sin\left(\omega t + \tfrac{\pi}{2}\right)$$

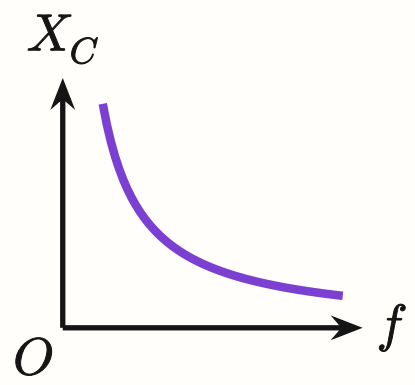
$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}, \qquad I_0 = \frac{V_0}{X_C}$$

current LEADS by $\frac{\pi}{2}$ · X_C: capacitive reactance

2 Graphs and phasor







- *i peaks a quarter period BEFORE v. $X_C \propto \frac{1}{f}$; DC (steady state): $X_C \to \infty$, BLOCKED once charged.*
- *Average power $= 0$: field energy stored in one quarter cycle is returned in the next.*

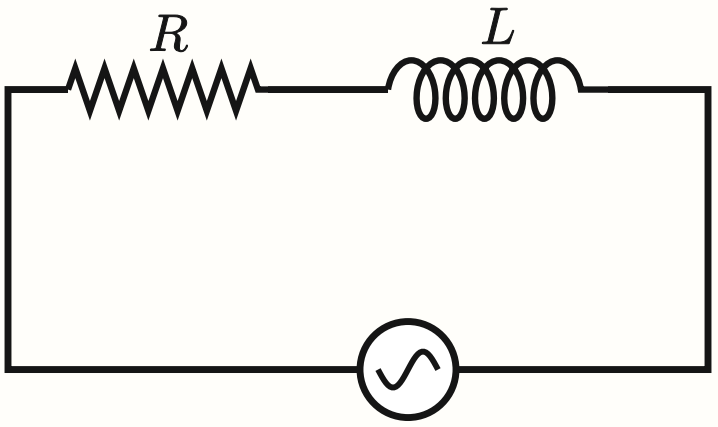
Example

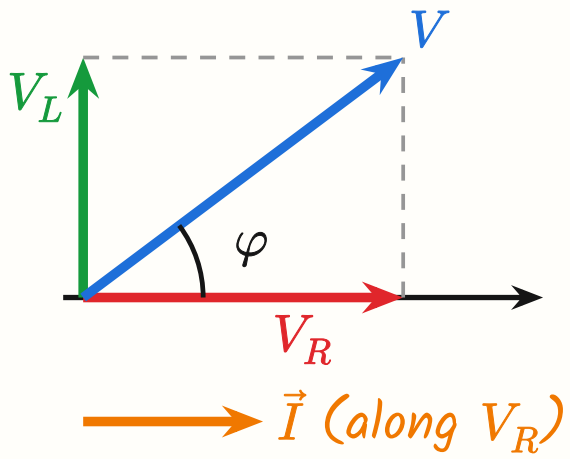
$C = 50\,\mu\text{F}$ *on* $220\,\text{V}$, $50\,\text{Hz}$: $X_C = \frac{1}{2\pi \times 50 \times 50 \times 10^{-6}} = 63.66\,\Omega$,
$I_{rms} = \frac{220}{63.66} = 3.46\,\text{A}$.

Series LR Circuit

R and an ideal L in SERIES across a sinusoidal source

1 Circuit and phasor diagram





2 Impedance and phase

$$Z = \sqrt{R^2 + X_L^2} \qquad \tan\varphi = \frac{X_L}{R}$$

$$I = \frac{V}{Z} \qquad V = \sqrt{V_R^2 + V_L^2}$$

SAME current in R and L (series) · $V_R = IR$ in phase with I, $V_L = IX_L$ leads I by $90°$ · $0 < \varphi < 90°$: current LAGS the source voltage

Example

$R = 40\,\Omega$, $L = 0.3\,\text{H}$, $\omega = 100\,\text{rad}\,\text{s}^{-1}$, $V = 200\,\text{V}$: $X_L = 30\,\Omega$, $Z = 50\,\Omega$, $I = 4\,\text{A}$, $V_R = 160\,\text{V}$, $V_L = 120\,\text{V}$, $\varphi = 36.87°$, $\cos\varphi = 0.8$.

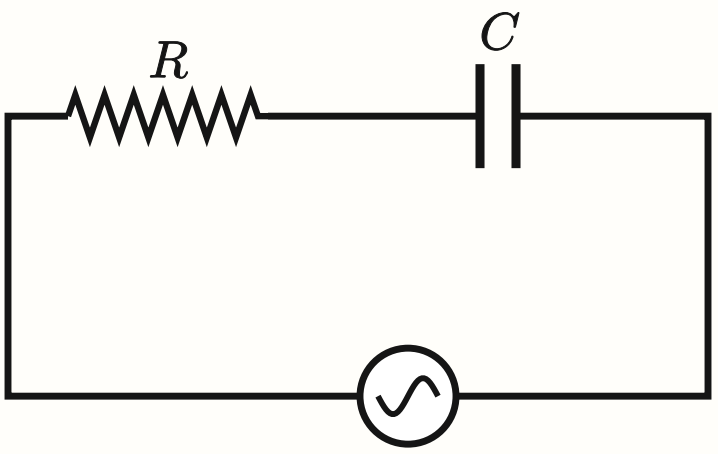
Golden points

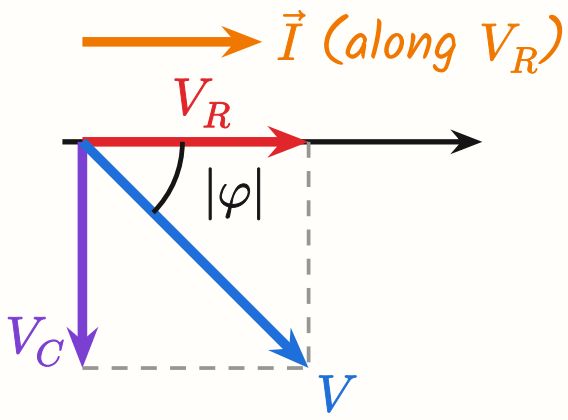
★ $160 + 120 \neq 200$: *rms voltages across series elements add as PHASORS, not as numbers.*

Series RC Circuit

R and an ideal C in SERIES across a sinusoidal source

1 Circuit and phasor diagram





2 Impedance and phase

$$Z = \sqrt{R^2 + X_C^2} \qquad \tan\varphi = -\frac{X_C}{R}$$

$$V = \sqrt{V_R^2 + V_C^2}$$

SAME current in R and C · $V_C = IX_C$ LAGS I by $90°$ · $-90° < \varphi < 0$: current LEADS the source voltage

Example

$R = 100\,\Omega$, $C = 10\,\mu\text{F}$, $\omega = 1000\,\text{rad}\,\text{s}^{-1}$, $V = 200\,\text{V}$: $X_C = 100\,\Omega$, $Z = 141.4\,\Omega$, $I = 1.414\,\text{A}$, $V_R = V_C = 141.4\,\text{V}$, *current LEADS by* $45°$.

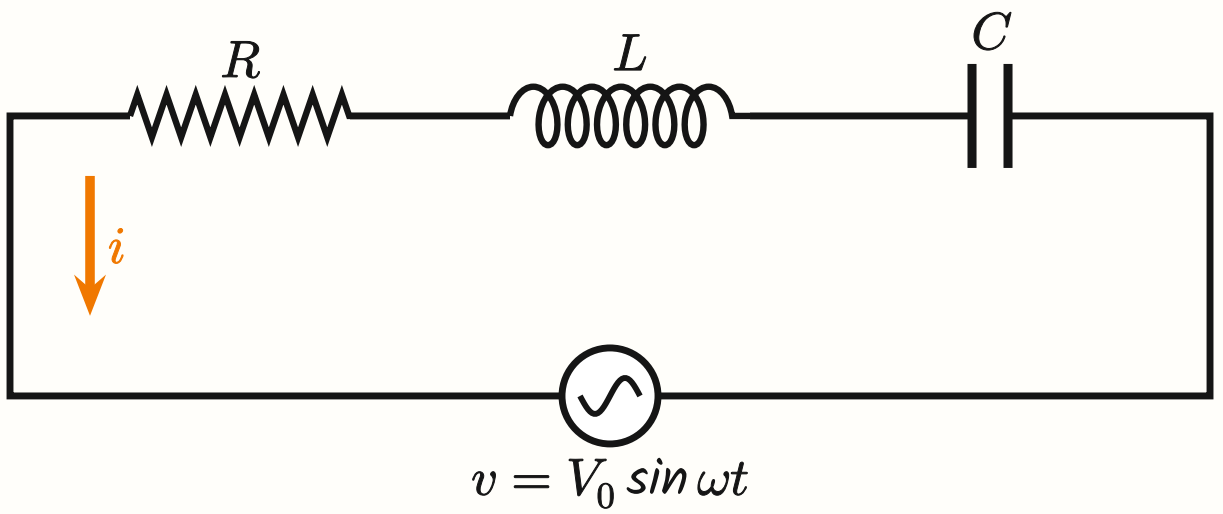
Golden points

★ *DC source (steady state): no current through the series RC branch; the capacitor takes the whole voltage.*

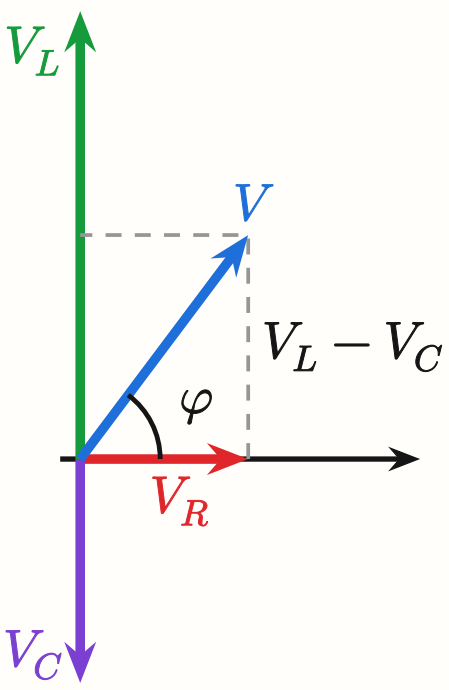
Series LCR Circuit

R, ideal L and ideal C in SERIES; φ = angle by which the source voltage LEADS the current

1 Circuit



2 Phasor diagram ($X_L > X_C$ drawn)



$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\tan\varphi = \frac{X_L - X_C}{R}$$

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$I = \frac{V}{Z}$ is the SAME in all three · V_L and V_C are OPPOSITE (180° apart), so they partly cancel · V_R along $\vec{I}$

- *$X_L > X_C$: $\varphi > 0$, current LAGS. $X_L < X_C$: $\varphi < 0$, current LEADS. $X_L = X_C$: $\varphi = 0$.*

LCR: Worked Example

series LCR, ideal L and C, sinusoidal source of fixed rms voltage

1 Numbers

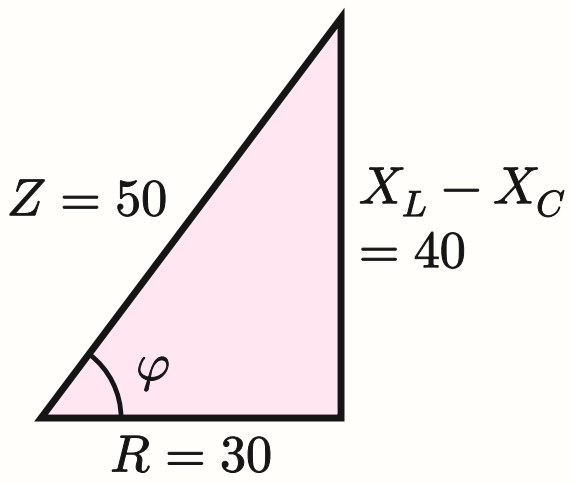
Example

$V = 100\,\text{V},\ R = 30\,\Omega,\ X_L = 80\,\Omega,\ X_C = 40\,\Omega$:
$Z = \sqrt{30^2 + 40^2} = 50\,\Omega,\ I = 2\,\text{A},\ V_R = 60\,\text{V},\ V_L = 160\,\text{V},$
$V_C = 80\,\text{V},\ \tan\varphi = \frac{4}{3},\ \varphi = 53.13^\circ$ *(current lags)*, $\cos\varphi = 0.6$.

Golden points

★ $V_L = 160\,\text{V}$ *EXCEEDS the source's* $100\,\text{V}$ *(check:* $\sqrt{60^2 + (160-80)^2} = 100\,\text{V}$*).*

2 Impedance triangle



- *Same shape as the voltage phasor triangle (every side* $\times I$*).*
- $\cos\varphi = \frac{R}{Z}$, $\sin\varphi = \frac{X_L - X_C}{Z}$.

3 Nature of the circuit

Condition ($\omega_0 = \frac{1}{\sqrt{LC}}$)	*behaves as*	*current*
$X_L > X_C$ $(\omega > \omega_0)$	*inductive*	*LAGS,* $\varphi > 0$
$X_L < X_C$ $(\omega < \omega_0)$	*capacitive*	*LEADS,* $\varphi < 0$
$X_L = X_C$ $(\omega = \omega_0)$	*purely resistive*	*in phase,* $\varphi = 0$

Series Resonance

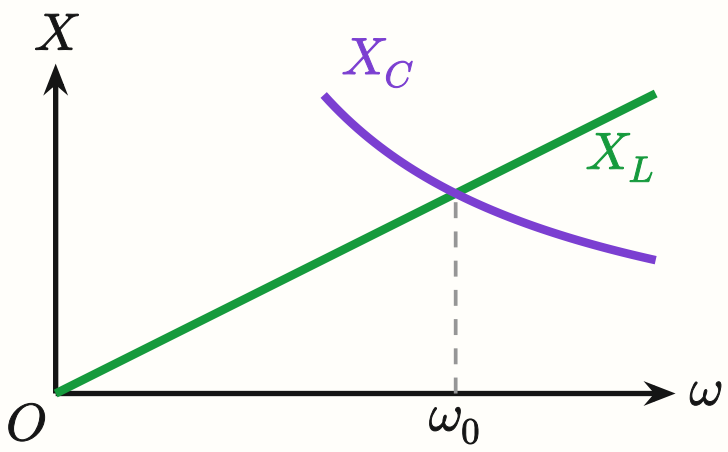
series LCR, fixed source rms voltage V, angular frequency ω VARIED

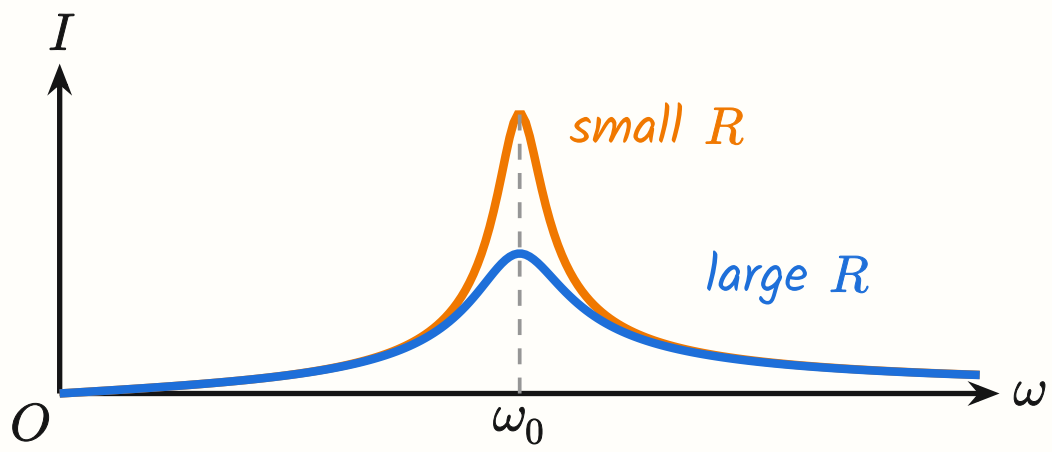
1 Condition

$$X_L = X_C \;\Rightarrow\; \omega_0 = \frac{1}{\sqrt{LC}}, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

at ω_0: $Z = R$ (MINIMUM), $I_{max} = \frac{V}{R}$, $\varphi = 0$, power factor 1 · $V_L = V_C$ and they cancel; $V_R = V$

2 Reactances and current against ω





X_L and X_C cross at ω_0. Current peaks at ω_0; smaller R: higher and SHARPER peak (drawn: R and $2R$).

Example

$L = 0.1\,\text{H}$, $C = 10\,\mu\text{F}$: $\omega_0 = \frac{1}{\sqrt{0.1\times10^{-5}}} = 1000\,\text{rad}\,\text{s}^{-1}$,

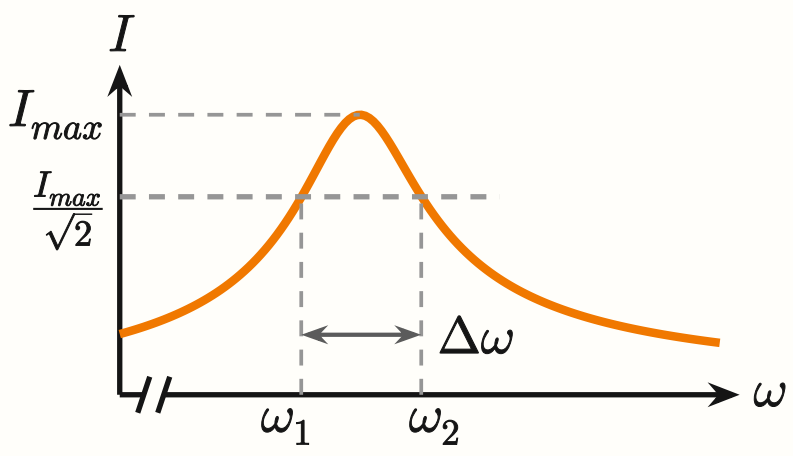
$f_0 = \frac{1000}{2\pi} = 159.2\,\text{Hz}$.

- *Used to TUNE a radio: C is varied until ω_0 equals the station's ω (acceptor circuit).*
- *Needs both L and C: an LR or RC circuit has no resonance.*

Q-factor and Bandwidth

series LCR, fixed source rms voltage, ω varied, $\omega_0 = 1/\sqrt{LC}$

1 Half-power frequencies



$$\omega_{1,2} = \mp\frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \omega_0^2}$$

$$\Delta\omega = \omega_2 - \omega_1 = \frac{R}{L}$$

at ω_1, ω_2: $I = \frac{I_{max}}{\sqrt{2}}$, power $= \frac{1}{2}P_{max}$ · EXACT for series LCR · $\omega_1\omega_2 = \omega_0^2$

2 Quality factor

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{\omega_0}{\Delta\omega}$$

dimensionless · at ω_0: $V_L = V_C = QV$ · large Q (small R): SHARP peak

Example

$L = 0.1\,\text{H}$, $C = 10\,\mu\text{F}$, $R = 10\,\Omega$, $V = 20\,\text{V}$: $\omega_0 = 1000$, $Q = \frac{1000\times0.1}{10} = 10$, $\Delta\omega = 100$, $\omega_1 = 951.2$, $\omega_2 = 1051.2$ (all $\text{rad}\,\text{s}^{-1}$), $I_{max} = 2\,\text{A}$, $V_L = V_C = 200\,\text{V}$.

Golden points

★ *ω varied, fixed source voltage, $Q > \frac{1}{\sqrt{2}}$: V_C peaks slightly BELOW ω_0 (at $\omega_0\sqrt{1 - \frac{1}{2Q^2}}$), V_L slightly ABOVE; only I and V_R peak exactly at ω_0.*

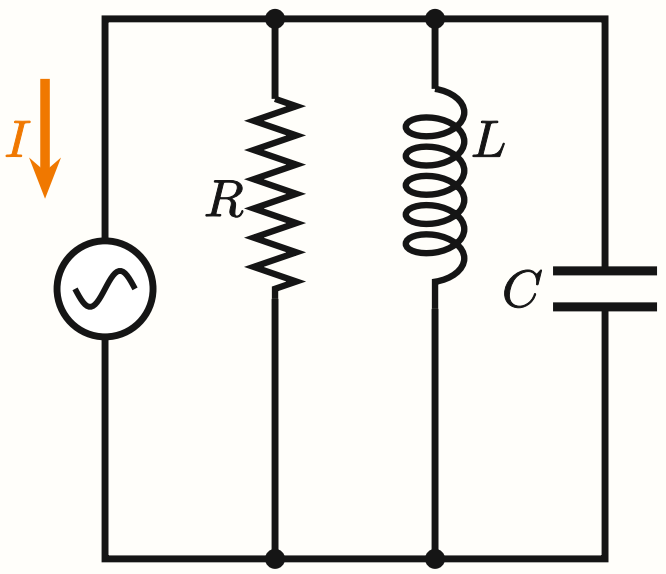
LC Series and Parallel Resonance

1 Series LC (no resistance)

$$Z = |X_L - X_C| \qquad \varphi = \pm 90^\circ$$

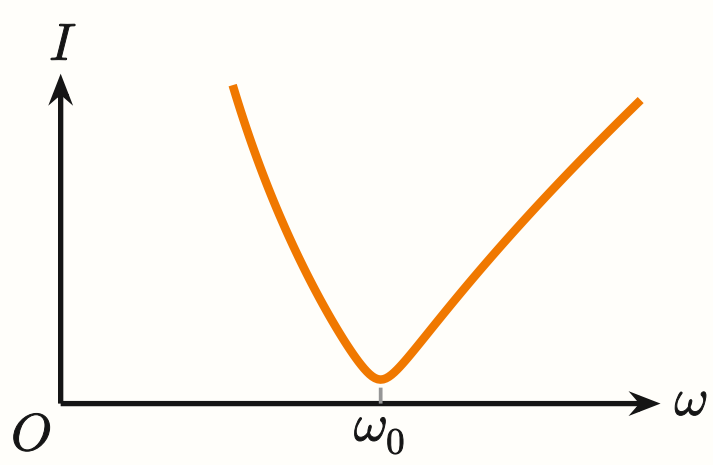
IDEAL L and C in series · $+90^\circ$ (lags) if $X_L > X_C$, -90° (leads) if $X_L < X_C$ · average power 0 · at ω_0: $Z \to 0$, the current is limited only by the small resistance always present

2 Parallel LCR



$$I = V\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

IDEAL R, L, C all in PARALLEL across the source, V rms · at $\omega_0 = \frac{1}{\sqrt{LC}}$: $I_L = I_C$, 180° apart, they cancel: line current MINIMUM $= \frac{V}{R}$, impedance MAXIMUM $= R$



- Line current is MINIMUM at ω_0 (series: maximum); $\omega \to 0$: ideal L shorts the source, $I \to \infty$.
- Ideal $L \parallel C$ alone ($R \to \infty$): line current 0, $Z \to \infty$ at ω_0: REJECTOR circuit. Series LCR: ACCEPTOR.

Example

$L = 0.1\,\mathrm{H}$, $C = 10\,\mu\mathrm{F}$, $R = 1000\,\Omega$ in parallel, $V = 100\,\mathrm{V}$: $\omega_0 = 1000\,\mathrm{rad\,s^{-1}}$, line current $0.1\,\mathrm{A}$, yet $I_L = I_C = \frac{100}{100} = 1\,\mathrm{A}$.

Power in AC Circuits

sinusoidal steady state; V, I rms; φ = phase difference between source voltage and current

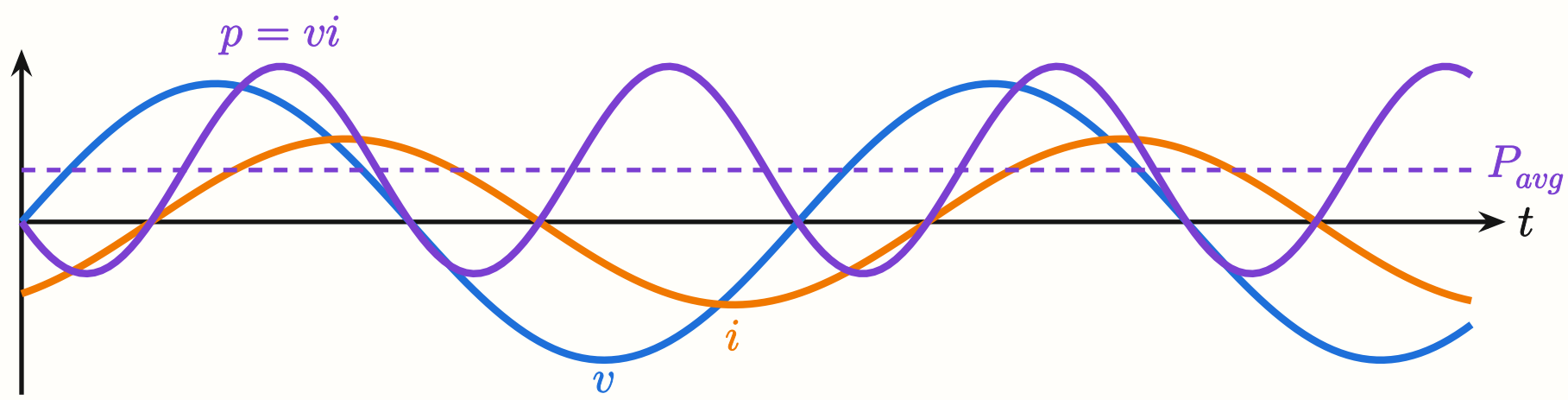
1 Instantaneous and average power

$$p = vi = V_0 I_0 \sin\omega t \, \sin(\omega t - \varphi)$$

$$P_{avg} = VI\cos\varphi = I^2 R$$

$\cos\varphi$ = POWER FACTOR $= \frac{R}{Z}$ (series circuit, R = total resistance)

2 p against t ($\varphi = 60°$ drawn)



p oscillates at 2ω about P_{avg}; $p < 0$: energy flows BACK to the source.

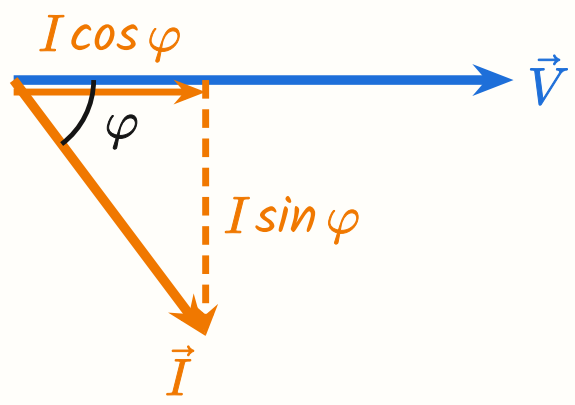
3 Power factor of each circuit

Circuit	$\cos\varphi$	P_{avg}
pure R	1	VI
ideal pure L or pure C	0	0
series LR, RC, LCR	$\frac{R}{Z}$	I^2R
series LCR at ω_0	1	$\frac{V^2}{R}$ (maximum)

Wattless Current, Power Triangle

sinusoidal steady state; V, I rms; numbers from the LCR example ($V = 100\,\mathrm{V}$, $I = 2\,\mathrm{A}$, $\varphi = 53.13°$)

1 Wattful and wattless components

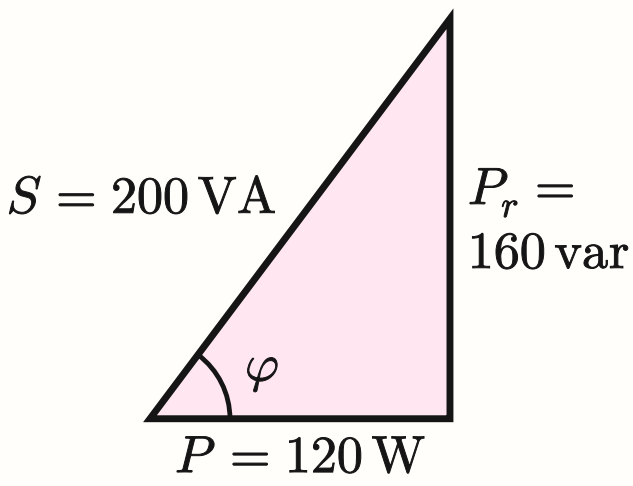


$$I_{wattful} = I\cos\varphi$$

$$I_{wattless} = I\sin\varphi$$

component IN PHASE with $\vec{V}$ gives the power; the one at $90°$ gives NONE (wattless) · example: $1.2\,\mathrm{A}$ and $1.6\,\mathrm{A}$

2 Apparent, true, reactive power



$$S = VI \quad (\mathrm{VA})$$

$$P = VI\cos\varphi \quad (\mathrm{W})$$

$$P_r = VI\sin\varphi \quad (\mathrm{var})$$

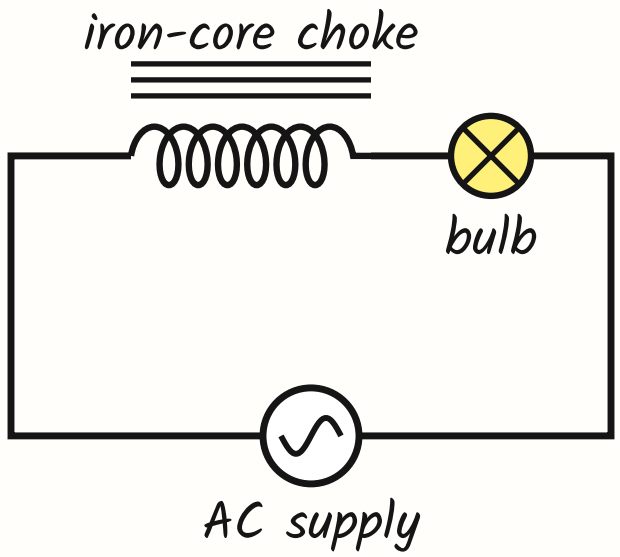
$S^2 = P^2 + P_r^2$ · example: $S = 100 \times 2 = 200\,\mathrm{VA}$, $P = 200 \times 0.6 = 120\,\mathrm{W} = I^2R$

- Low power factor: MORE current for the same power, so more I^2R loss in the supply lines.
- Inductive loads (motors) are corrected with a capacitor in PARALLEL, which raises $\cos\varphi$ towards 1.

Choke Coil

AC (sinusoidal) supply; choke = coil of LARGE inductance and SMALL resistance r

1 Idea



$$P_{avg} = I^2 r \approx 0$$

$$\cos\varphi = \frac{r}{\sqrt{r^2 + \omega^2 L^2}} \approx 0$$

$\omega L \gg r$: the choke limits the AC current through its REACTANCE while wasting almost no power · a series resistor doing the same job wastes $I^2 R$

2 Bulb on a higher-voltage supply

Example

Bulb rated $100\,\text{V}$, $50\,\text{W}$ on a $200\,\text{V}$, $50\,\text{Hz}$ supply. Bulb: $I = \frac{50}{100} = 0.5\,\text{A}$, $R_b = \frac{100^2}{50} = 200\,\Omega$. **Series resistor:** drop $100\,\text{V}$: $R = 200\,\Omega$, wasted $0.5^2 \times 200 = 50\,\text{W}$. **Ideal choke:** $Z = \frac{200}{0.5} = 400\,\Omega$, $X_L = \sqrt{400^2 - 200^2} = 346.4\,\Omega$, $L = \frac{346.4}{2\pi \times 50} = 1.10\,\text{H}$, wasted ≈ 0.

Golden points

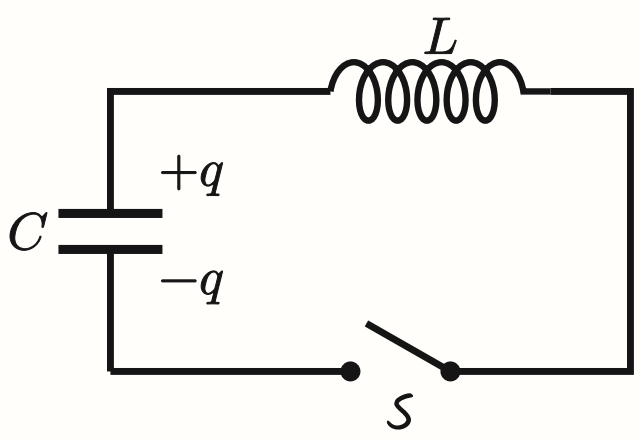
★ Voltages: bulb $100\,\text{V}$, choke $0.5 \times 346.4 = 173.2\,\text{V}$; their sum is not $200\,\text{V}$ (phasors at $90°$).

- Steady DC: only the small r limits the current, so a choke is useful for AC only. Laminated soft-iron core: large L, small core loss.

LC Oscillations

ideal L and C (no resistance); C charged to q_0, switch S closed at $t = 0$; $\omega_0 = \dfrac{1}{\sqrt{LC}}$

1 Circuit and equation



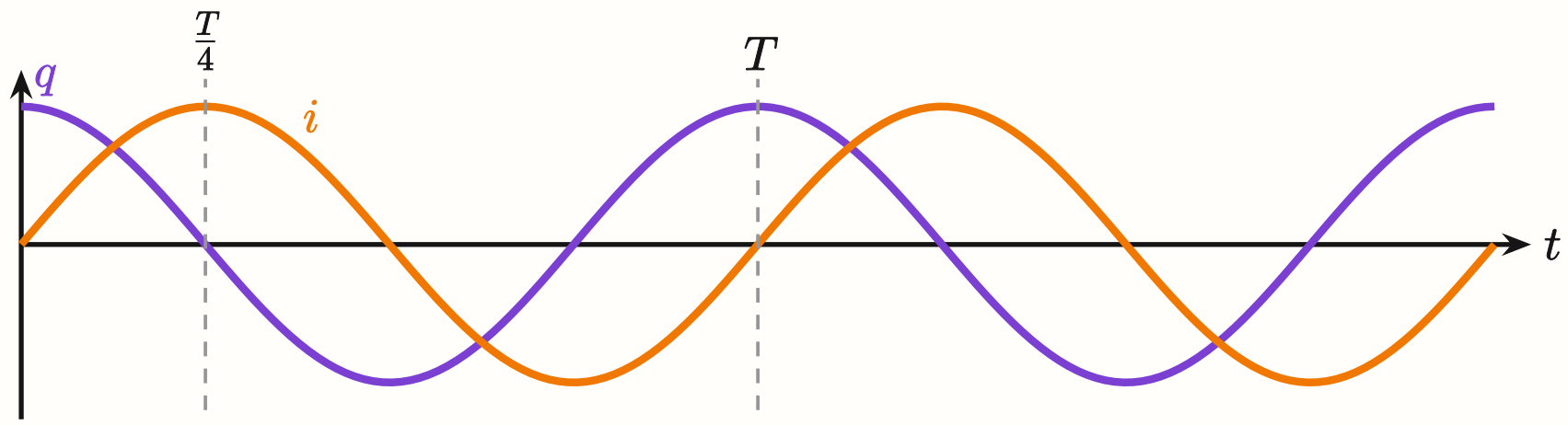
$$L\frac{d^2q}{dt^2} + \frac{q}{C} = 0$$

$$q = q_0 \cos \omega_0 t$$

$$i = -\frac{dq}{dt} = \omega_0 q_0 \sin \omega_0 t$$

i = discharge current · $I_{max} = \omega_0 q_0$ · $f_0 = \frac{1}{2\pi\sqrt{LC}}$

2 q and i against time



i is maximum when $q = 0$, zero when $q = \pm q_0$: a quarter period ($\frac{T}{4}$) apart.

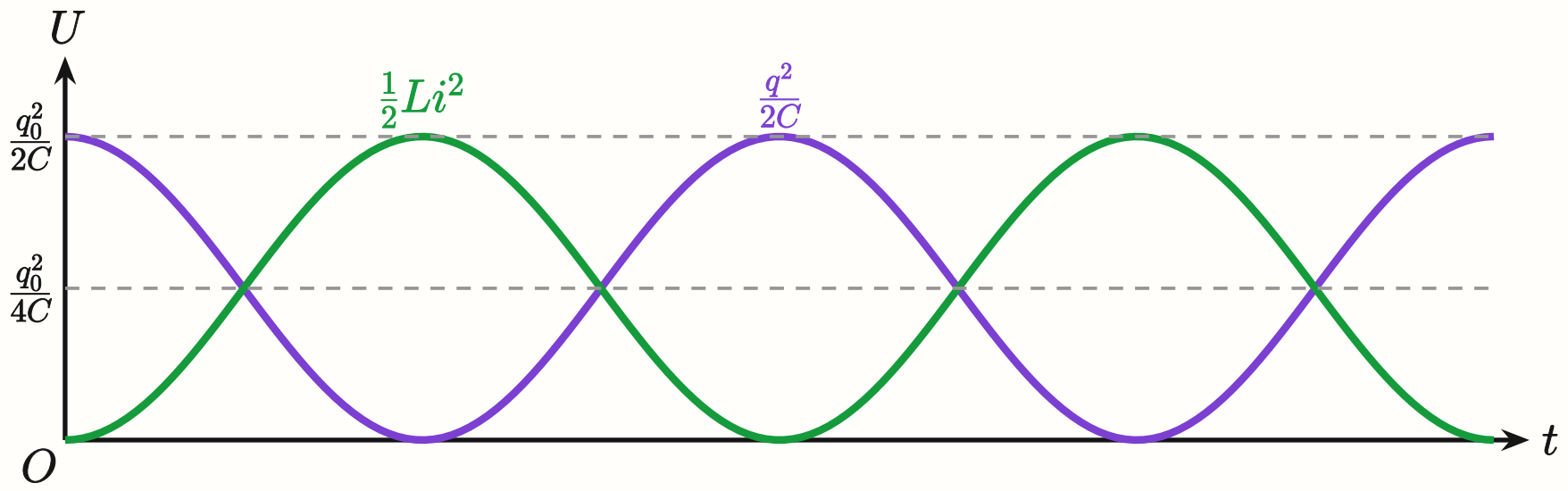
Example

$L = 20\,\text{mH}$, $C = 50\,\mu\text{F}$, $q_0 = 100\,\mu\text{C}$: $\omega_0 = \frac{1}{\sqrt{10^{-6}}} = 1000\,\text{rad s}^{-1}$, $f_0 = 159.2\,\text{Hz}$, $T = 6.28\,\text{ms}$, $I_{max} = 0.1\,\text{A}$.

LC Oscillations: Energy

ideal L and C (no resistance, no radiation); C charged to q_0 at $t = 0$

1 Energy swings between C and L



$$\frac{q^2}{2C} + \frac{1}{2}Li^2 = \frac{q_0^2}{2C}$$

total energy CONSTANT · $U_C = \frac{q_0^2}{2C}\cos^2\omega_0 t$, $U_L = \frac{q_0^2}{2C}\sin^2\omega_0 t$ · each repeats every HALF period

- U_C is ELECTRIC (field between the plates), U_L MAGNETIC (field in the coil). Each oscillates at $2f_0$; its average over a cycle is $\frac{q_0^2}{4C}$.

Example

Same circuit as before ($L = 20\,\text{mH}$, $C = 50\,\mu\text{F}$, $q_0 = 100\,\mu\text{C}$): energy $\frac{q_0^2}{2C} = \frac{10^{-8}}{10^{-4}} = 10^{-4}\,\text{J}$; check $\frac{1}{2}LI_{max}^2 = \frac{1}{2}\times 0.02 \times 0.1^2 = 10^{-4}\,\text{J}$.

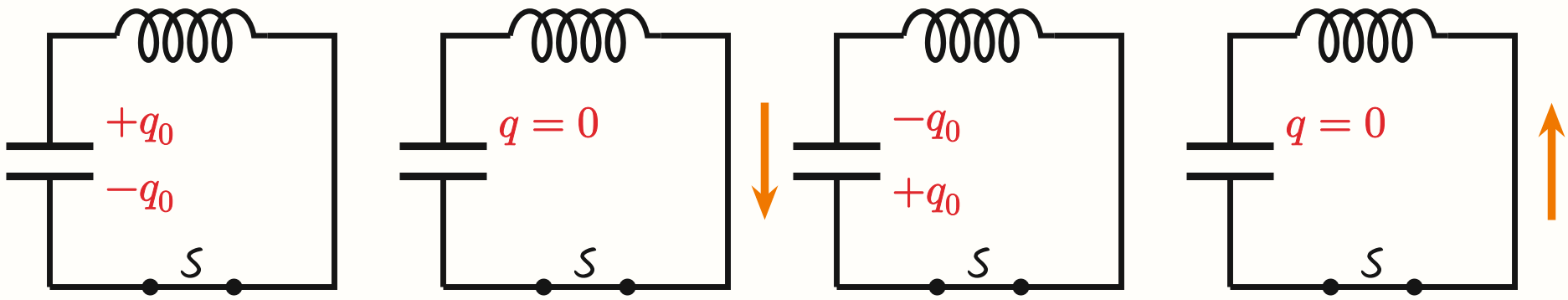
Golden points

★ $|i| = \omega_0\sqrt{q_0^2 - q^2}$ (like $v = \omega\sqrt{A^2 - x^2}$); $U_C = U_L$ when $q = \frac{q_0}{\sqrt{2}}$, first at $t = \frac{T}{8}$.

LC Oscillations and SHM

ideal L and C (no resistance); C charged to q_0 (upper plate $+$), switch S closed at $t = 0$

1 Four instants of one cycle



$t = 0$: all in C — $\frac{T}{4}$: all in L — $\frac{T}{2}$: all in C — $\frac{3T}{4}$: all in L

Discharge current CLOCKWISE, maximum at $\frac{T}{4}$; plates REVERSED at $\frac{T}{2}$; current maximum ANTICLOCKWISE at $\frac{3T}{4}$.

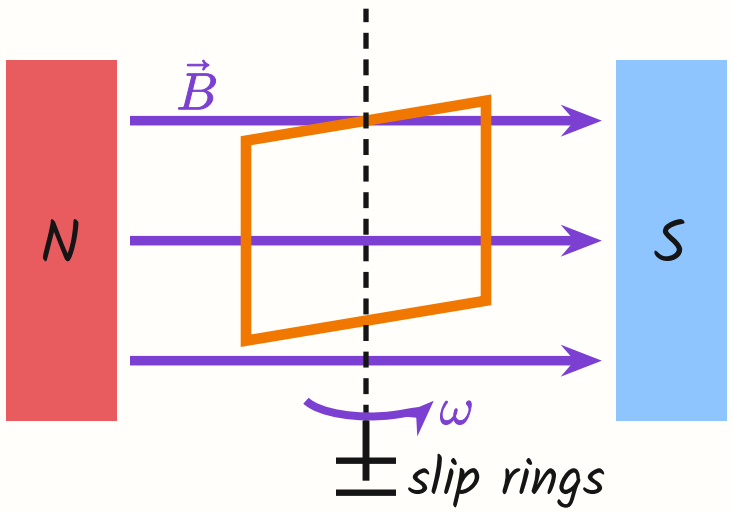
2 Analogy with SHM (spring-block)

LC circuit	spring-block (SHM)
charge q, rate $\frac{dq}{dt}$	displacement x, velocity $v = \frac{dx}{dt}$
inductance L, $\frac{1}{C}$	mass m, spring constant k
$\frac{q^2}{2C}$ (electric)	$\frac{1}{2}kx^2$ (potential)
$\frac{1}{2}Li^2$ (magnetic)	$\frac{1}{2}mv^2$ (kinetic)
$\omega_0 = \frac{1}{\sqrt{LC}}$	$\omega = \sqrt{k/m}$

- Equation $\frac{d^2q}{dt^2} = -\omega_0^2 q$ has the SHM form $\frac{d^2x}{dt^2} = -\omega^2 x$: same solutions, same energy swapping.
- Real circuit (resistance R): oscillations DECAY (damped, like friction in SHM); oscillatory only if $R < 2\sqrt{L/C}$.

AC Generator

1 Coil rotating in a uniform field



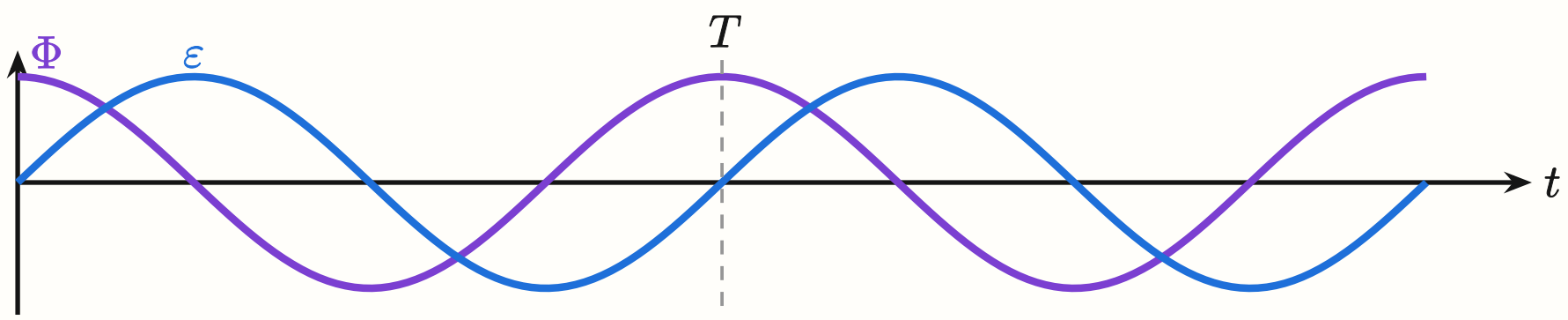
$$\varepsilon = NBA\omega \sin \omega t$$

$$\varepsilon_0 = NBA\omega$$

N turns, area A, axis $\perp$ UNIFORM $\vec{B}$, CONSTANT ω · $t = 0$: plane of the coil $\perp \vec{B}$ · from Faraday's law, $\Phi = BA\cos\omega t$ per turn

- ε is MAXIMUM when the plane of the coil is PARALLEL to $\vec{B}$ (flux zero), ZERO when it is $\perp \vec{B}$ (flux maximum).
- AC frequency = revolutions per second; doubling ω doubles BOTH ε_0 and the frequency.
- The torque on the induced current OPPOSES the rotation (Lenz): mechanical work $\rightarrow$ electrical energy.

2 Flux and EMF against time



$\varepsilon = -N\frac{d\Phi}{dt}$: ε is zero when Φ is extreme; it LAGS Φ by a quarter period.

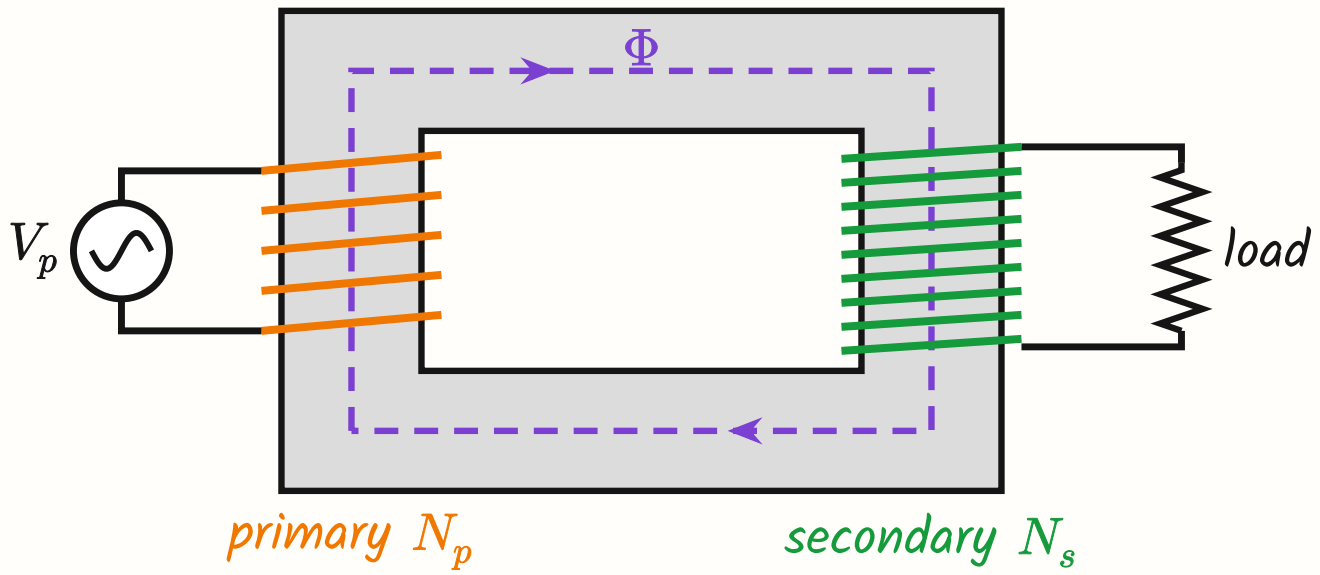
Example

$N = 50$, $A = 0.02\,\mathrm{m}^2$, $B = 0.5\,\mathrm{T}$, $50\,\mathrm{rev\,s^{-1}}$ ($\omega = 314.16\,\mathrm{rad\,s^{-1}}$): $\varepsilon_0 = 157.1\,\mathrm{V}$, *rms* $111.1\,\mathrm{V}$.

Transformer

mutual induction: an AC in the primary makes a changing flux in a common soft-iron core

1 Construction



2 Ideal transformer

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

IDEAL: ALL the flux links both coils, coil resistances zero, no core losses ($V_p I_p = V_s I_s$) · rms values, sinusoidal AC · $N_s > N_p$: step-UP (voltage up, current DOWN); $N_s < N_p$: step-DOWN

- *Same $\frac{d\Phi}{dt}$ in every turn, so the EMF PER TURN is the same in both coils.*
- *Steady DC in the primary: constant flux, NO output (except at switching). A transformer does not increase POWER.*

Example

$N_p = 500$, $N_s = 2500$, $V_p = 220\,\text{V}$: $V_s = 1100\,\text{V}$. *Load current* $I_s = 0.2\,\text{A}$: $I_p = 1\,\text{A}$; *power* $220\,\text{W}$ *on both sides.*

Transformer: Efficiency and Losses

1 Efficiency

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

real transformer: $\eta < 100\%$ *(large ones* ≈ 95–99%*)* · $P_{in} = P_{out}$ + *losses*

Example

Input $1000\,\mathrm{W}$, *output* $950\,\mathrm{W}$: $\eta = 95\%$, *loss* $50\,\mathrm{W}$ *(heat).*

2 Losses and remedies

Loss	remedy
copper (I^2R in the windings)	thick copper wire
eddy currents in the core	LAMINATED core (insulated sheets)
hysteresis (core magnetised back and forth)	soft iron (narrow hysteresis loop)
flux leakage	coils wound one over the other

Golden points

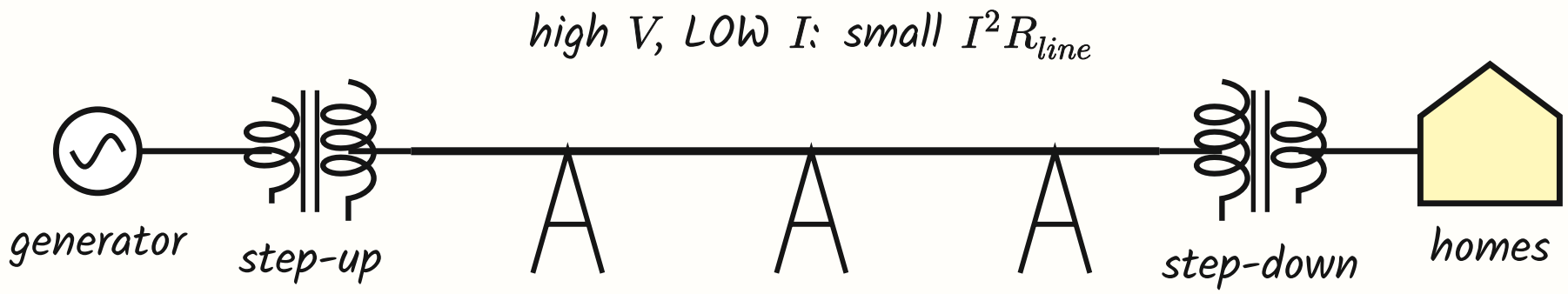
★ Copper loss $\propto I^2$: it grows with the LOAD. Core (iron) loss = eddy + hysteresis: nearly the same at any load (fixed supply voltage and frequency).

★ No moving parts, so large transformers reach ≈ 95–99%; the lost energy appears as HEAT (oil or air cooling).

Power Transmission

power P SENT into the line at rms voltage V, power factor $\cos\varphi$, line resistance R_{line}, ideal transformers

1 Generator to home



2 Line loss

$$I = \frac{P}{V\cos\varphi} \qquad P_{loss} = I^2 R_{line} = \frac{P^2 R_{line}}{V^2\cos^2\varphi}$$

V = rms voltage at which the power is sent · same P, $V \times k$: current $\div k$, loss $\div k^2$ · low $\cos\varphi$: MORE current and loss

Example

$22\,\text{kW}$ *sent at unity power factor into a line of* $0.5\,\Omega$*: at* $220\,\text{V}$*,* $I = 100\,\text{A}$*, loss* $5000\,\text{W}$*; at* $22\,\text{kV}$*,* $I = 1\,\text{A}$*, loss* $0.5\,\text{W}$*.*

- *AC is used for distribution because transformers step it UP at the power station and DOWN near the consumer with very little loss.*
- *A step-up transformer raises V and LOWERS I by the same factor: the power itself is not increased.*

Special Cases (I)

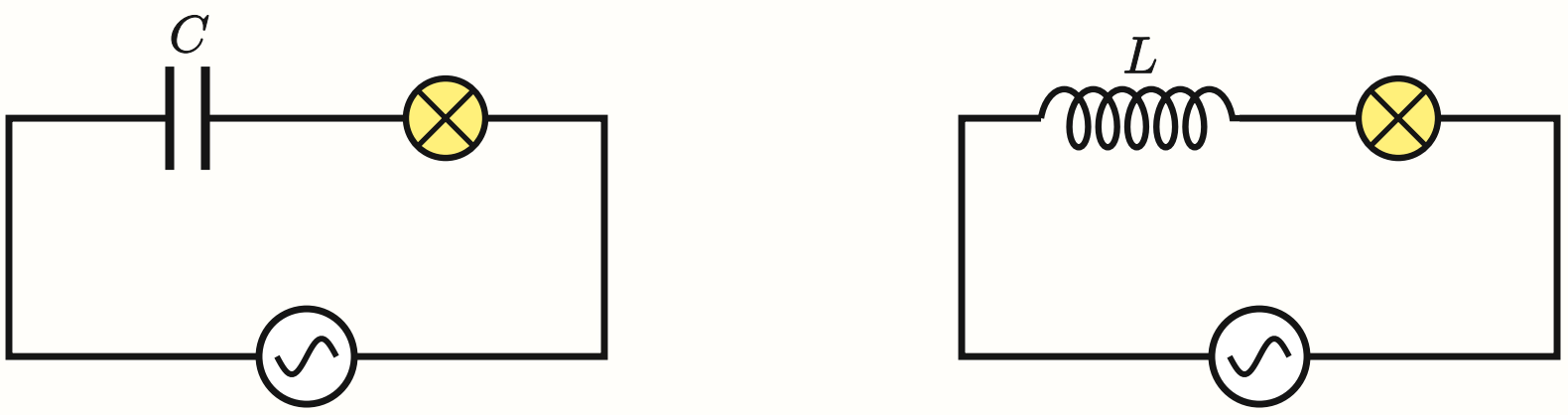
1 DC source, steady state

- Ideal inductor: $X_L = 0$ at $f = 0$, a plain WIRE; a coil of resistance r on a DC source ε: $I = \frac{\varepsilon}{r}$.
- Capacitor in series with the DC source (no other path): once charged, NO current; it holds the full source voltage.

Example

A coil draws $2.5\,\mathrm{A}$ from $100\,\mathrm{V}$ DC: $r = 40\,\Omega$. On $100\,\mathrm{V}$ rms, $50\,\mathrm{Hz}$ it draws less: $Z = \sqrt{r^2 + X_L^2} > r$.

2 Bulb in series, source rms voltage fixed



- With C: raise f or insert a dielectric (C up): X_C falls, bulb BRIGHTER.
- With L: raise f or insert a soft-iron rod (L up): X_L rises, bulb DIMMER.

3 Series LCR at extreme frequencies

- $\omega \to 0$: $X_C \to \infty$, $I \to 0$, C takes the whole source voltage.
  $\omega \to \infty$: $X_L \to \infty$, $I \to 0$, L takes it.

Special Cases (2)

1 Real coil (inductor with resistance r)

$$V_{coil} = I\sqrt{r^2 + X_L^2}, \quad \tan\theta = \frac{X_L}{r}$$

voltage across the coil LEADS the current by $\theta < 90^\circ$ (not 90°) · the coil takes power $I^2 r$

2 Meter readings in a series LCR

Example

Rms voltmeter readings $V_R = 40\,\text{V}$, $V_L = 80\,\text{V}$, $V_C = 50\,\text{V}$ (ideal L, C): source $V = \sqrt{40^2 + (80-50)^2} = 50\,\text{V}$, power factor $\frac{40}{50} = 0.8$.

Golden points

★ *Ideal L, C: readings $V_L = V_C$ mean resonance; then V_R = source voltage.*

3 Maximum power with R varied

$$P = \frac{V^2 R}{R^2 + X^2} \text{ is maximum at } R = |X|$$

$$P_{max} = \frac{V^2}{2|X|}, \quad \cos\varphi = \frac{1}{\sqrt{2}}$$

series circuit, $X = X_L - X_C \neq 0$ FIXED, only R varied, rms V fixed

Example

$V = 100\,\text{V}$, $X = 50\,\Omega$: *best* $R = 50\,\Omega$, $P_{max} = \frac{100^2}{100} = 100\,\text{W}$.

Standard Results

1 One-line results

Quantity	Result	Condition
rms, sine	$\frac{V_0}{\sqrt{2}}$	sinusoidal
mean, sine	$\frac{2V_0}{\pi}$; 0	positive half cycle; full cycle
Reactances	$X_L = \omega L,\ X_C = \frac{1}{\omega C}$	ideal L, C
Impedance	$\sqrt{R^2 + (X_L - X_C)^2}$	series LCR
Phase	$\tan\varphi = \frac{X_L - X_C}{R}$	$\varphi > 0$: current lags
Resonance	$\omega_0 = \frac{1}{\sqrt{LC}},\ Z = R$	series LCR, $X_L = X_C$
Q-factor	$\frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$	series LCR
Bandwidth	$\Delta\omega = \frac{R}{L}$	half-power points, series LCR
Power	$VI\cos\varphi = I^2 R$	rms, sinusoidal, average
Power factor	$\cos\varphi = \frac{R}{Z}$	series circuit
Wattless	$I\sin\varphi$	component $\perp \vec{V}$
DC + AC	$\sqrt{I_1^2 + \frac{I_2^2}{2}}$	$i = I_1 + I_2\sin\omega t$
LC	$\omega_0 = \frac{1}{\sqrt{LC}},\ I_{max} = \omega_0 q_0$	no resistance
Generator	$\varepsilon_0 = NBA\omega$	axis $\perp$ uniform $\vec{B}$
Transformer	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$	ideal
Line loss	$\frac{P^2 R_{line}}{V^2\cos^2\varphi}$	P sent at rms V

Summary

$$V_{rms} = \frac{V_0}{\sqrt{2}} \bullet Z = \sqrt{R^2 + (X_L - X_C)^2} \bullet P = VI\cos\varphi$$

sinusoidal · series LCR · rms values, average power

Common Traps

1 Exam traps

Avoid these

- ✗ Adding rms voltages in series: $V = V_R + V_L + V_C$ (no: $V = \sqrt{V_R^2 + (V_L - V_C)^2}$).
- ✗ Mains $220\,\mathrm{V}$ is the PEAK (no: rms; peak $311\,\mathrm{V}$).
- ✗ $X_L = fL$ (no: $X_L = \omega L = 2\pi f L$).
- ✗ Mean of a sine is $\frac{2V_0}{\pi}$ always (only over a POSITIVE half cycle; over a full cycle 0).
- ✗ Pure L or C draws no current (no: current flows, but the AVERAGE power is zero).
- ✗ Power factor $= \frac{X}{Z}$ (no: $\cos\varphi = \frac{R}{Z}$).
- ✗ $Q = \frac{\omega L}{R}$ at any ω (no: at ω_0).
- ✗ Bandwidth $\frac{R}{L}$ is in Hz (no: rad s^{-1}; in Hz $\Delta f = \frac{R}{2\pi L}$).
- ✗ A step-up transformer increases power (no: ideal, $V_p I_p = V_s I_s$; the current steps DOWN).
- ✗ A moving-coil meter shows the rms of an AC (no: the mean, zero for sine).
- ✗ At ω_0 the voltages across L and C are zero (no: equal and OPPOSITE, each QV).
- ✗ A capacitor passes steady DC (no: it blocks it once charged).

Golden points

- ★ First ask: series or parallel? which phasor is the reference? rms or peak? at ω or at ω_0?

Questions: JEE Main Level (I)

sinusoidal sources, ideal L and C, steady state, rms values unless stated

Q1 · JEE Main level · MCQ

$v = 311\sin(100\pi t)$ V (t in s). Rms value and frequency: (A) $220\,\text{V}$, $50\,\text{Hz}$ (B) $311\,\text{V}$, $50\,\text{Hz}$ (C) $220\,\text{V}$, $100\,\text{Hz}$ (D) $156\,\text{V}$, $50\,\text{Hz}$

Solution: $V_{rms} = \frac{311}{\sqrt{2}} = 219.9 \approx 220\,\text{V}$; $f = \frac{100\pi}{2\pi} = 50\,\text{Hz}$.

Answer: (A)

Q2 · JEE Main level · Numerical

An ideal inductor of $\frac{1}{\pi}\,\text{H}$ is connected to $220\,\text{V}$, $50\,\text{Hz}$. Rms current (in A)?

Solution: $X_L = 2\pi \times 50 \times \frac{1}{\pi} = 100\,\Omega$; $I = \frac{220}{100} = 2.2\,\text{A}$.

Answer: 2.2

Q3 · JEE Main level · MCQ

An ideal capacitor draws $2\,\text{A}$ from $220\,\text{V}$, $50\,\text{Hz}$. From $220\,\text{V}$, $100\,\text{Hz}$ it draws: (A) $1\,\text{A}$ (B) $2\,\text{A}$ (C) $4\,\text{A}$ (D) $8\,\text{A}$

Solution: $X_C \propto \frac{1}{f}$ halves, so $I = \frac{V}{X_C}$ doubles: $4\,\text{A}$.

Answer: (C)

Q4 · JEE Main level · MCQ

Series LCR: $R = 6\,\Omega$, $X_L = 12\,\Omega$, $X_C = 4\,\Omega$, $V = 100\,\text{V}$. Average power: (A) $1000\,\text{W}$ (B) $600\,\text{W}$ (C) $800\,\text{W}$ (D) $360\,\text{W}$

Solution: $Z = \sqrt{36 + 64} = 10\,\Omega$, $I = 10\,\text{A}$, $\cos\varphi = 0.6$, $P = 100 \times 10 \times 0.6 = 600\,\text{W} = I^2 R$.

Answer: (B)

Questions: JEE Main Level (2)

sinusoidal sources, ideal L and C, steady state, rms values unless stated

Q5 · JEE Main level · Numerical

Series LCR: $L = 0.5\,\text{H}$, $C = 8\,\mu\text{F}$, $R = 25\,\Omega$. Find ω_0 (in $\text{rad}\,\text{s}^{-1}$) and the Q-factor.

Solution: $\omega_0 = \frac{1}{\sqrt{0.5 \times 8 \times 10^{-6}}} = 500$; $Q = \frac{\omega_0 L}{R} = \frac{500 \times 0.5}{25} = 10$.

Answer: 500, 10

Q6 · JEE Main level · MCQ

A transformer ($N_p = 1000$) steps $220\,\text{V}$ down to $11\,\text{V}$, supplying $2\,\text{A}$ to a resistive load at 90% efficiency (voltage ratio = turns ratio). N_s, I_p: (A) 50, $0.1\,\text{A}$ (B) 50, $0.111\,\text{A}$ (C) 20, $0.111\,\text{A}$ (D) 50, $0.09\,\text{A}$

Solution: $N_s = 1000 \times \frac{11}{220} = 50$; $P_{in} = \frac{22}{0.9} = 24.44\,\text{W}$, $I_p = 0.111\,\text{A}$.

Answer: (B)

Q7 · JEE Main level · Numerical

$i = 3 + 4\sin\omega t$ A. Rms value (in A)?

Solution: $I_{rms} = \sqrt{3^2 + 4^2/2} = \sqrt{17} = 4.12\,\text{A}$.

Answer: 4.12

Q8 · JEE Main level · MCQ

A $60\,\text{V}$, $30\,\text{W}$ bulb (resistor) on $220\,\text{V}$, $50\,\text{Hz}$ with an ideal inductor in series. L: (A) $0.38\,\text{H}$ (B) $1.35\,\text{H}$ (C) $1.40\,\text{H}$ (D) $0.67\,\text{H}$

Solution: $I = \frac{30}{60} = 0.5\,\text{A}$, $R_b = 120\,\Omega$, $Z = 440\,\Omega$,

$X_L = \sqrt{440^2 - 120^2} = 423.3\,\Omega$, $L = \frac{423.3}{100\pi} = 1.35\,\text{H}$.

Answer: (B)

Questions: JEE Advanced Level (I)

sinusoidal sources, ideal L and C, steady state, rms values unless stated

A1 · JEE Advanced level · Multi-correct

Series LCR: $R = 20\,\Omega$, $L = 0.2\,\mathrm{H}$, $C = 5\,\mu\mathrm{F}$, $40\,\mathrm{V}$ rms source, ω variable: (A) the current is maximum at $\omega = 1000\,\mathrm{rad\,s^{-1}}$ (B) at that ω the rms voltage across L is $400\,\mathrm{V}$ (C) the half-power bandwidth is $100\,\mathrm{rad\,s^{-1}}$ (D) at that ω the average power is $160\,\mathrm{W}$

Solution: $\omega_0 = \frac{1}{\sqrt{0.2\times5\times10^{-6}}} = 1000$; $I_{max} = \frac{40}{20} = 2\,\mathrm{A}$, $V_L = 2\times1000\times0.2 = 400\,\mathrm{V}$; $\Delta\omega = \frac{R}{L} = 100$; $P = \frac{V^2}{R} = 80\,\mathrm{W}$, not 160.

Answer: (A), (B), (C)

A2 · JEE Advanced level · Numerical

A coil draws $4\,\mathrm{A}$ from a $120\,\mathrm{V}$ DC source and $2.4\,\mathrm{A}$ from a $120\,\mathrm{V}$ rms, $50\,\mathrm{Hz}$ source. Find (i) r and L (ii) the power factor and power on AC (iii) the capacitance put in SERIES to make the power factor 1 at $50\,\mathrm{Hz}$, and then the current, the power and the rms voltage across C.

Solution: (i) $r = \frac{120}{4} = 30\,\Omega$; $Z = \frac{120}{2.4} = 50\,\Omega$, $X_L = \sqrt{50^2 - 30^2} = 40\,\Omega$, $L = \frac{40}{100\pi} = 0.127\,\mathrm{H}$. (ii) $\cos\varphi = \frac{30}{50} = 0.6$, $P = 2.4^2\times30 = 172.8\,\mathrm{W}$. (iii) $X_C = X_L = 40\,\Omega$: $C = \frac{1}{100\pi\times40} = 79.6\,\mu\mathrm{F}$; now $Z = r$: $I = 4\,\mathrm{A}$, $P = 480\,\mathrm{W}$, $V_C = 4\times40 = 160\,\mathrm{V}$.

Answer: (i) $30\,\Omega$, $0.127\,\mathrm{H}$ (ii) 0.6, $172.8\,\mathrm{W}$ (iii) $79.6\,\mu\mathrm{F}$, $4\,\mathrm{A}$, $480\,\mathrm{W}$, $160\,\mathrm{V}$

Questions: JEE Advanced Level (2)

peak value V_0; ideal L and C; capacitor charged before the switch is closed at $t = 0$

A3 · JEE Advanced level · Matching

List I (periodic voltage of peak V_0): (P) sine (Q) square $\pm V_0$ (R) symmetric triangular $\pm V_0$ (S) half-wave rectified sine. List II (rms value): (1) $\frac{V_0}{\sqrt{3}}$ (2) $\frac{V_0}{2}$ (3) V_0 (4) $\frac{V_0}{\sqrt{2}}$

Solution: $\overline{\sin^2} = \frac{1}{2}$; square: $v^2 = V_0^2$ always; triangle: $\overline{x^2}$ of a linear ramp $= \frac{V_0^2}{3}$; half-wave: $\frac{1}{2} \times \frac{V_0^2}{2}$ (zero for half the period).

Answer: P–4, Q–3, R–1, S–2

A4 · JEE Advanced level · Numerical

A $2\,\mu\text{F}$ capacitor charged to $50\,\text{V}$ is connected through switch S to an ideal $5\,\text{mH}$ inductor at $t = 0$. Find (i) ω_0 (ii) maximum current (iii) the first instant (in μs) when the energy is shared equally (iv) the current when $q = \frac{q_0}{2}$.

Solution: $q_0 = CV = 100\,\mu\text{C}$; $\omega_0 = \frac{1}{\sqrt{5\times10^{-3}\times2\times10^{-6}}} = 10^4\,\text{rad s}^{-1}$;

$I_{max} = \omega_0 q_0 = 1\,\text{A}$; $\cos^2\omega_0 t = \frac{1}{2}$ first at $\omega_0 t = \frac{\pi}{4}$; $t = 78.5\,\mu\text{s}$;

$|i| = \omega_0\sqrt{q_0^2 - \frac{q_0^2}{4}} = 0.866\,\text{A}$.

Answer: (i) $10^4\,\text{rad s}^{-1}$ (ii) $1\,\text{A}$ (iii) 78.5 (iv) $0.866\,\text{A}$

Golden points

★ rms of ANY periodic waveform $= \sqrt{\text{mean of } v^2 \text{ over one period}}$.

★ Ideal LC: $|i| = \omega_0\sqrt{q_0^2 - q^2}$; energy shared equally first at $\frac{T}{8}$.

Questions: JEE Advanced Level (3)

unity power factor, purely resistive line; series circuit with ideal L and a variable C

A5 · JEE Advanced level · Numerical

A load takes $10\,\mathrm{kW}$ at unity power factor through a line of total resistance $0.5\,\Omega$ (no line reactance). The line loss must be 1% of the power delivered. Find (i) the line current (ii) the rms load voltage and the sending-end voltage (iii) the loss if the sending voltage were only $220\,\mathrm{V}$, with the same $10\,\mathrm{kW}$ delivered.

Solution: (i) $I^2 \times 0.5 = 100\,\mathrm{W}$: $I = \sqrt{200} = 14.14\,\mathrm{A}$. (ii) $V_{load} = \frac{10000}{14.14} = 707.1\,\mathrm{V}$; the line drop $IR = 7.07\,\mathrm{V}$ is in phase (resistive line, unity power factor): $V_{send} = 714.2\,\mathrm{V}$. (iii) $220I - 0.5I^2 = 10000$: $I = 220 - \sqrt{28400} = 51.48\,\mathrm{A}$ (smaller root), loss $0.5 \times 51.48^2 = 1325\,\mathrm{W}$, about 13%.

Answer: (i) $14.14\,\mathrm{A}$ (ii) $707.1\,\mathrm{V}$, $714.2\,\mathrm{V}$ (iii) $\approx 1325\,\mathrm{W}$

A6 · JEE Advanced level · Numerical

$R = 100\,\Omega$ and $L = 1\,\mathrm{H}$ in series with a variable C, on $200\,\mathrm{V}$ rms at $\omega = 100\,\mathrm{rad\,s^{-1}}$. (i) C (in $\mu\mathrm{F}$) for maximum current, and that current? (ii) With C halved: current and phase?

Solution: (i) $X_C = X_L = 100\,\Omega$: $C = \frac{1}{\omega^2 L} = 100\,\mu\mathrm{F}$, $I = \frac{200}{100} = 2\,\mathrm{A}$. (ii) $X_C = 200\,\Omega$, $Z = \sqrt{100^2 + 100^2} = 141.4\,\Omega$, $I = 1.414\,\mathrm{A}$, $\tan\varphi = \frac{100-200}{100} = -1$: current LEADS by $45°$.

Answer: (i) 100, $2\,\mathrm{A}$ (ii) $1.414\,\mathrm{A}$, leads by $45°$

Golden points

★ Line loss $\propto \frac{1}{V^2}$ at fixed power SENT (fixed R_{line} and $\cos\varphi$): the reason for high-voltage transmission.